

Bridging the Trust Gap: Conformal Prediction for AI-based Intracranial Hemorrhage Detection

Peter K. Ngum^{1,2,3} • Christopher G. Filippi, MD^{4,5}

Peter K. Ngum is a PhD candidate in clinical neuroscience at the Turku Brain Injury Center, Finland. His research explores explainable and interpretable multimodal data harmonization approaches for traumatic brain injury. Dr Ngum is particularly interested in federated analysis and responsible AI practices, including uncertainty quantification.



Christopher “Risto” G. Filippi, MD, is a professor of radiology at the University of Toronto, chief of pediatric neuroradiology at the Hospital for Sick Children, and a senior associate research scientist at the Sick Kids Institute. His research focuses on AI, the translation of advanced MRI techniques, glymphatic flow, and adolescent brain development.



Integrating artificial intelligence (AI) into radiology holds significant promise to streamline diagnostic workflows and improve patient outcomes. Among the most clinically impactful applications is the automated detection of intracranial hemorrhage (ICH)—an urgent, high-stakes task in which rapid and accurate diagnosis can substantially influence patient management. To date, deep learning–based models have demonstrated strong performance in identifying various ICH subtypes, including intraparenchymal, intraventricular, subdural, epidural, and subarachnoid hemorrhage (1). Despite these advances, radiologists and referring clinicians may remain cautious about adopting AI models more widely. The underlying challenge is trust: To integrate AI seamlessly into patient care, there is a need for models that provide not only accurate predictions but also transparent assessments of model uncertainty, particularly for borderline or atypical cases.

In their article in this issue of *Radiology: Artificial Intelligence*, Gamble and Faghani et al (2) propose a strategy to improve the trustworthiness of an ICH detection model. They couple the state-of-the-art YOLO (You Only Look Once) version 8 object detection architecture (3) with a form of conformal prediction, a statistical framework that quantifies formal uncertainty. Instead of producing only a single class label or an uncalibrated probability, a conformal method outputs a set

of plausible labels at a user-specified confidence level (4). This approach can mimic how uncertainty is expressed in clinical settings, for example, by listing a differential diagnosis with multiple possibilities rather than stating one diagnosis with unwarranted certainty.

Bridging the Trust Gap in Clinical AI

Accurate detection of ICH on CT is crucial in emergency settings. Overlooking a subtle epidural hematoma can be life-threatening, whereas overcalling a hemorrhage can lead to unnecessary interventions. Most deep learning models generally produce only a single-label output, sometimes accompanied by confidence scores that may be prone to miscalibration, especially in out-of-distribution scenarios such as motion artifacts, metallic implants, or unusual hemorrhage appearances. If a model is not explicitly trained on these edge cases, its “confidence” values could be misleadingly high even when uncertain.

Gamble and Faghani et al address this issue by adopting Mondrian conformal prediction (MCP), a variant of conformal prediction that calibrates each class separately to enable more granular handling of heterogeneous ICH subtypes (5). MCP calibrates the model’s raw confidence across multiple hemorrhage classes, learning how these values relate to ground truth labels in a calibration dataset. When an unseen image is evaluated, the method checks how the model’s output compares to this calibration distribution. If the test image behaves “like” those calibration samples with strong confidence in subdural hemorrhage, for instance, the algorithm includes subdural hemorrhage in the predicted label set. If it also resembles samples with a decent probability of being epidural hemorrhage, the system may return both subdural hemorrhage and epidural hemorrhage as potential diagnoses, or it may flag the image as “challenging,” cueing the radiologist to examine that scan more closely.

Study Overview and Findings

Gamble and Faghani et al trained a YOLO version 8 model on 491 noncontrast head CT volumes from the publicly available CQ500 dataset (6). Three senior radiologists independently labeled hemorrhages in these scans, leading to the exclusion of 29 volumes that contained no hemorrhage. Of the remaining 462 positive volumes, 146 cases were classified as “definite” (where all radiologists agreed) and 150 cases as “challenging” (where disagreement arose). Additionally, 166 volumes were excluded due to ambiguous labeling. The authors then allocated subsets of these definite volumes for training, tuning, calibration, and testing. The challenging set was reserved for evaluating whether the MCP-augmented model could identify ambiguous scenarios.

Author affiliations, funding, and conflicts of interest are listed at the end of this article.

See also article by Gamble and Faghani et al in this issue.

Radiology: Artificial Intelligence 2025; 7(2):e250032 • <https://doi.org/10.1148/ryai.250032> • Content codes: **AI** **NR** • © RSNA, 2025

This copy is for personal use only. To order printed copies, contact reprints@rsna.org

The key step involved generating Mondrian groups: For each hemorrhage subtype—and its complement, that is, “no hemorrhage” of that subtype—the YOLO version 8 model’s confidence scores were sorted and recorded in calibration. During inference, bounding boxes for each subtype were selected, and the model derived conformal scores to decide which hemorrhage classes to include in the final set-based prediction.

Their results are promising. The hybrid model showed high overall performance, identified ambiguous cases, and performed well in external testing. On the definite test dataset, the model attained an F1 score of 0.92 for hemorrhage classification, which places the model near levels reported for similar tasks. The model correctly flagged 95.3% of the challenging scans as uncertain. Rather than forcing an incorrect single-label assignment, the model effectively communicated that multiple interpretations were possible. The authors further applied their approach to the multi-institutional RSNA Brain CT Hemorrhage Challenge dataset to demonstrate that the method generalizes beyond the CQ500 images, albeit with some performance variation that reflected real-world heterogeneity (7).

By explicitly signaling which cases require closer human review, this AI framework could prevent certain missed diagnoses and reduce overconfidence-driven errors. From a workflow standpoint, it offers a safety net: Straightforward scans receive high-confidence predictions, while boundary or out-of-distribution examples are placed in a “review required” queue.

Clinical Significance and Practical Considerations

1. Calibrated confidence: In emergency radiology, time is critical. With conformal methods, clinicians can tune the coverage probability (eg, 90%, 95%, or 99%) according to their risk tolerance. An urgent trauma center might opt for a very high sensitivity threshold (to avoid misses) and accept more false positives. A center dealing with routine scans might tighten specificity requirements.

2. Interpretation and visualization: YOLO version 8 provides bounding boxes around detected regions of hemorrhage, facilitating direct visualization of suspected abnormal sites. When combined with MCP, a bounding box can carry a set of probable classes. This approach mirrors: “This region could be a thin subdural hematoma or perhaps an epidural hematoma; correlation with additional slices is needed.”

3. Integration with clinical workflow: By automatically flagging challenging scans, the model could reinforce a form of triage; easy cases can be confirmed rapidly. In contrast, a second opinion or senior radiologist review is prioritized for ambiguous cases. This synergy of human expertise plus AI-driven uncertainty detection could bolster patient safety while improving efficiency, embodying a form of human-centric AI.

Limitations and Future Directions

Although their findings are compelling, several practical factors warrant further investigation. First, the calibration dataset focused heavily on volumes with unanimous radiologist agree-

ment, which may underrepresent the full spectrum of ambiguous presentations. Incorporating uncertain or atypical volumes into calibration might strengthen the robustness of uncertainty estimates. Second, real-world populations vary substantially in demographics, imaging acquisition parameters, and comorbid conditions. Ensuring that training and calibration data are demographically and clinically diverse is essential to prevent biased performance or calibration drift. Third, many radiologists rely on volumetric interpretation rather than section-by-section analysis. Extending MCP to full three-dimensional bounding boxes might yield more holistic insights. Fourth, a feedback loop that systematically analyzes flagged challenging cases—and the eventual clinical outcomes—could refine calibration thresholds or model parameters over time, ensuring the system adapts to shifts in practice or imaging technology. Finally, conformal prediction can be applied across various radiologic tasks, from characterizing lung nodules to identifying subtle tumor recurrences. Studies that replicate these methods in other domains will help confirm their general feasibility and impact.

In summary, Gamble and Faghani et al illustrate how a careful integration of MCP with a modern YOLO-based detection network can meaningfully bridge the trust gap in automated ICH detection. Rather than providing a single point prediction, the MCP-augmented model delivers a set of plausible diagnoses or flags the case as challenging—a more clinically intuitive approach that parallels the differential diagnosis process. By helping radiologists see when and why the system is uncertain, such methods foster safer interaction between AI and humans.

Author affiliations:

¹ Turku Brain Injury Center, TYKS Turku University Hospital, Turku, Finland

² Carey Business School, John Hopkins University, 100 International Dr, Baltimore, MD 21202

³ Center for Equitable AI Health, Baltimore, Md

⁴ Department of Radiology, SickKids Research Institute, Toronto, Canada

⁵ Department of Radiology at the Hospital for Sick Children, University of Toronto, Toronto, Canada

Received January 8, 2025; revision requested January 15; revision received February 19; accepted February 25.

Address correspondence to: P.K.N. (email: pngum1@jhu.edu).

Funding: Authors declared no funding for this work.

Acknowledgment: The authors gratefully acknowledge Anastasios Angelopoulos, a PhD candidate in electrical engineering and computer science at the University of California, Berkeley, for his valuable contributions to this commentary, specifically his insights on the exchangeability assumption for the calibration dataset.

Disclosures of conflicts of interest: P.K.N. Founder and CEO of Center for Equitable AI Health and Pan African AI Health Initiative (neither entity has provided direct support for this manuscript or study). C.G.F. Paid by NAMS to interpret brain MRI for clinical trials for a device company (screening for presence of acute stroke); Data Safety Monitoring Board for Protocol Title: A Phase III Randomized Trial of Repeated Supraselective Intraarterial Cerebral infusion (SIACI) of Bevacizumab (Avastin) with Temozolomide and Radiation Compared to Temozolomide and Radiation Alone in Newly Diagnosed Glioblastoma (GBM) (Principal Investigator: John Boockvar, MD); member of Scientific Advisory Council; secretary of Peds MR User Group of ISMRM; stock/stock options in Avicenna.ai, minority (< 1%) stockholder.

References

1. Wang X, Shen T, Yang S, et al. A deep learning algorithm for automatic detection and classification of acute intracranial hemorrhages in head CT scans. *Neuroimage Clin* 2021;32:102785.
2. Gamble C, Faghani S, Erickson BJ. Applying conformal prediction to a deep learning model for intracranial hemorrhage detection to improve trustworthiness. *Radiol Artif Intell* 2025;7(2):e240032.

3. Sohan M, Sai Ram T, Rami Reddy CV. A review on YOLOv8 and its advancements. In: Jacob JJ, Piramuthu S, Falkowski-Gilski P, eds. Data Intelligence and Cognitive Informatics. ICDICI 2023. Algorithms for Intelligent Systems. Springer, 2024.
4. Angelopoulos AN, Bates S. Conformal prediction: a gentle introduction. Found Trends Mach Learn 2023;16(4):494–591.
5. Boström H, Johansson U, Löfström T. Mondrian conformal predictive distributions. In: Lars C, Zhiyuan L, Giovanni C, Khuong An N, eds. Proceedings of the Tenth Symposium on Conformal and Probabilistic Prediction and Applications. Proc Mach Learn Res 2021;152:24–38. <https://proceedings.mlr.press/v152/bostrom21a.html>.
6. Chilamkurthy S, Ghosh R, Tanamala S, et al. Deep learning algorithms for detection of critical findings in head CT scans: a retrospective study. Lancet 2018;392(10162):2388–2396.
7. Flanders AE, Prevedello LM, Shih G, et al. Construction of a machine learning dataset through collaboration: The RSNA 2019 brain CT hemorrhage challenge. Radiol Artif Intell 2020;2(3):e190211.